Homework 9

Due: *Monday, July 26th at 11:59pm EDT*

All homeworks are due the Monday after their release at 11:59pm EDT on Gradescope.

Please do not include any identifying information about yourself in the handin, including your Banner ID.

Be sure to fully explain your reasoning and show all work for full credit. We recommend checking out the sample proofs on the course website to see the level of rigor for which we're looking.

Problem 1

For each of the following pairs of events, identify whether they are independent. Justify why or why not by using the fact that event A and B are independent if $P(A) = P(A|B)$.

a. When flipping a fair coin three times:

- The first coin comes up tails (T)
- There is a run of exactly two heads (that is, there are two, but not three, heads flipped in a row)

b. When generating a 0/1 string of length 5:

- The 3rd bit is a 1
- There are at least two 0s

Problem 2

The 14 moons of Neptune work at Neptune's Intergalactic Inc. Each of them works at their respective desk. Triton, one of Neptune's worker moons, wants to prank Neso, another one of Neptune's worker moons. Knowing Neso hates disorder, Triton convinces everyone in the office to switch desks when Neso walks out of the room. The prank is only successful if there is exactly one moon per desk after the jumble and the new arrangement is not identical to the previous one.

a. If each moon chooses a desk uniformly and independently (potentially choosing their own desk), provide an expression for the probability the prank is successful (that is, each desk has exactly one person and at least one person has moved desks)?

b. Now, consider a case in which all desks are arranged in a circle, and the moons only have time to move to a desk directly to the left or right of their current desk. What is the probability the prank is successful? (Every moon moves desks this time.)

c. Neso is taking a little longer to return to the office—the moons will move to any desk exactly 2 to the right or left of their original position. In this case, what is the probability that the prank is successful?

Problem 3

Neptune is given the following equation:

$$x_1 + x_2 + x_3 + x_4 = 75$$

where each x_i must be non-negative.

a. Count the number of solutions to this equation.
Hint: Use Donuts and Separators!

b. Now suppose we require a solution with x_1 and x_3 strictly positive. Count the number of solutions under this new constraint.

c. Let us now assume that each x_i for $i \in [1, 4]$ takes some value in the range $[0, 75]$ uniformly at random. Given that x_1 and x_3 are strictly positive, what is the probability that $x_1 + x_2 + x_3 + x_4 = 75$?

Mind Bender (Extra Credit)

Neptune is working on a research paper but its only local planetary library has made it so that each member can only take out a single book. In front of Neptune is an aisle of n different books relevant to its paper, where n is a positive integer.

Neptune doesn't have enough time to go through every book, so it wants to select a book **uniformly at random**. In other words, Neptune looks to find a scheme that will guarantee that it selects the i^{th} book with probability $\frac{1}{n}$, where $1 \leq i \leq n$.

Neptune has some constraints. The planet is too tired to make multiple trips up and down the aisle, so it looks to go down the aisle only once. Furthermore, Neptune can only hold one book at a time.

Come up with and prove a scheme that allows Neptune to walk away from the end of the aisle with the i^{th} book with probability $\frac{1}{n}$.